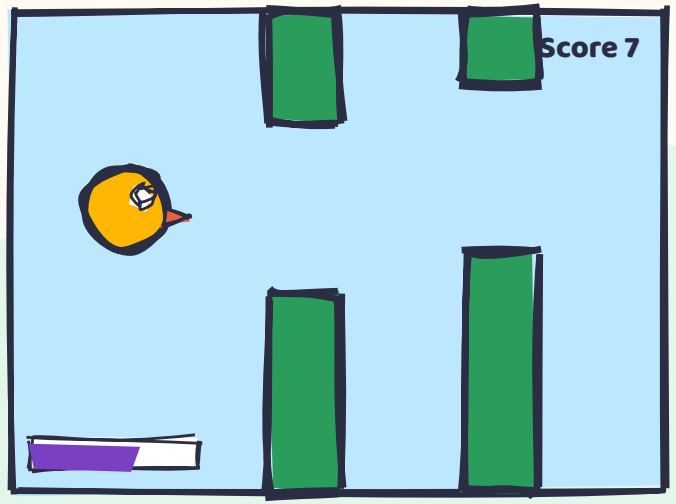
Young Innovators · Class 29

p5.js: Flappy Voice Game

Hum to fly. Stay quiet to fall.

Mic level | Gravity | Pipes & score



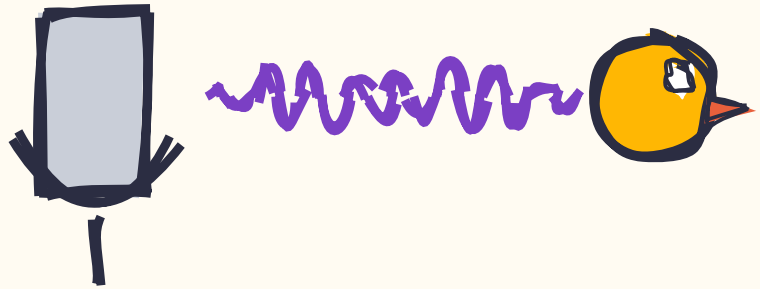
Did you know?
Voice-controlled games help players who can't use their hands. Some gamers play whole games with sounds and breath.

Let's think...

Can you play a game with no hands?

Hands-free

Imagine a game you control only with your voice: hum louder to fly higher, go quiet to sink. Fun for everyone, and a real way to include players who can't press buttons.



Shout at it?

Too hard!

Use the mic!

Today: you build **Flappy Voice** in the p5.js web editor: a bird that flies when you make noise.

Learning objectives

By the end of this class I can...

1	**Sound → numbers:** explain how a microphone turns sound into a **level** from 0 to 1	**Proof:** my game shows a live volume bar
2	**Game physics:** program **gravity**, **velocity** and a voice-powered **lift**	**Proof:** my bird falls and rises smoothly when I hum
3	**Full game:** add moving **pipes**, **collisions** and a **score**, then playtest	**Proof:** a classmate scored at least 5 using only their voice

LEARN

Objective 1 · Sound → numbers · video

Video time!



Microphones, Loudspeakers & Headphones

FuseSchool · about 4 minutes · youtu.be/dzDye6mi-6M

Watch for...

1. What part of a microphone **vibrates**?
2. How does sound become an **electrical signal**?
3. How is a loudspeaker like a microphone in **reverse**?

Scan to watch

After the video: Hum quietly, then loudly. What would a microphone 'see' differently?

LEARN

Objective 1 · Sound → numbers

A microphone turns sound into numbers

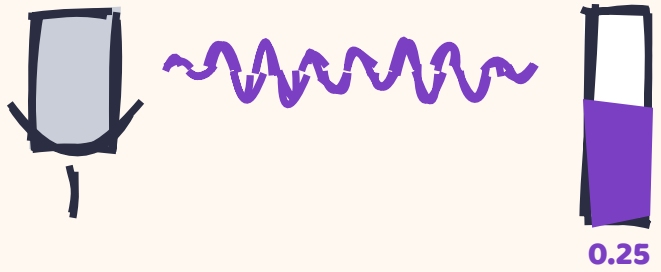
The idea

Sound is air **vibrating**. Inside a microphone, a thin **diaphragm** vibrates too and makes a tiny changing voltage.

The computer measures it thousands of times a second. p5.sound gives us the **level**: **0** = silence, about **0.02** = quiet room, **0.1 to 0.3** = humming, up to **1** = very loud.

mic.getLevel() → one number per frame.

Example: a drum skin



Put rice on a drum and hit it: the rice jumps. Louder hit, higher jump. The diaphragm is a tiny drum, and the level is how high the rice jumps.

LEARN

Objective 2 · Game physics

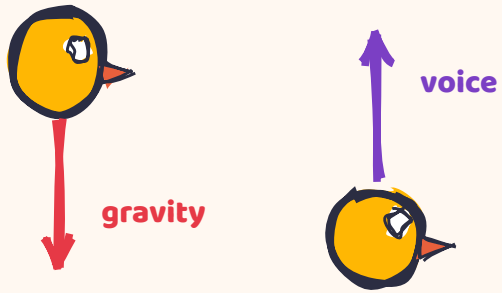
Gravity, velocity, lift

The idea

Each frame:
vy = vy + gravity (always pulling down, e.g. +0.4)
if level > threshold: vy = vy − level × power (voice lifts)
y = y + vy (move)

Velocity (vy) is speed with a direction: + = falling, − = rising. Adding gravity every frame makes falling **speed up**, like real life. **constrain()** stops it going crazy fast.

Example: throwing a ball up



Throw a ball up: it slows, stops, then falls faster and faster. Gravity changes its velocity a little every moment. Our code does the same, 60 times a second.

LEARN

Objective 3 · Full game

Pipes, collisions and score

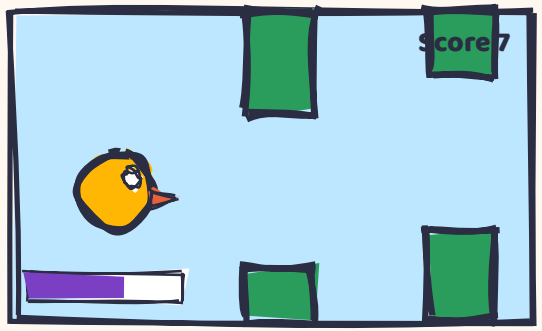
The idea

Every 90 frames, add a **pipe** at the right edge with a random gap height. Each frame, move all pipes left by 3 px.

Collision: the bird hits a pipe if it is **horizontally inside** the pipe AND **outside** the gap. Also: hitting the floor or ceiling.

Score +1 when a pipe passes the bird. Remove pipes that leave the screen to keep the game fast.

Example: a conveyor belt



Airport bags on a belt: new ones appear at one end, move along, and drop off the other. Our pipes are bags on a conveyor belt!

Real world

Voice and sound control



Accessible gaming
Voice and breath controllers for disabled gamers



Singing games
Karaoke apps score your pitch and volume



Voice assistants
Wake word + loudness detection



Breathing apps
Spirometer games help children with lung exercises



Who else could your voice game help? What would they need it to do differently?

Design for everyone
Making a game playable without hands is called accessible design. It often makes games better for everyone.

LEARN

Quick quiz

Game physics

Question	Answer
1. Silence gives mic level about…?	
2. vy = 3. Is the bird going up or down?	
3. Why add gravity EVERY frame?	
4. Two conditions for hitting a pipe?	
5. Why remove pipes that leave the screen?	

Answers

Cover! 1: close to 0 · 2: down (+ is down on screen) · 3: so falling speeds up like real gravity · 4: inside the pipe's x AND outside the gap · 5: keep the game fast

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human flappy
2. Our setup
3. Bird physics with your voice
4. Pipes, crashes and score
5. Polish and playtest
6. Playtest notes
7. Game designer challenges

MAKE

Class game

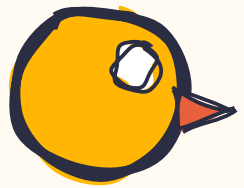
Human flappy

How to play

1. One student is the **bird**, standing by the wall. The class is the **microphone**.
2. When the class hums, the bird rises up the wall (hand sliding up). Silence: the bird sinks slowly.
3. Teacher holds two rulers as a 'gap' and moves it. The class must hum to steer the bird through!
4. Try: very loud = rocket! Who controls the bird best?

Then discuss

Was it easier with short hums or one long hum? What does that tell you about the 'power' setting?



Big idea

Small changes in power and gravity change how a game feels.

Build it

Our setup



editor.p5js.org
Free web editor, no install



Laptop microphone
Or a headset mic



Account (optional)
To save and share sketches



p5.sound
Already included in new sketches



Sprites (optional)
Your own bird drawing



Headphones
So games don't get too loud

Browsers only allow the mic after you click: our code calls userStartAudio() on the first click.

Noise
In a noisy class every bird flies! Use headset mics, or raise the threshold.

MAKE

Objective 2 · Game physics · Build step 1 of 3

Bird physics with your voice

```
let mic, y = 300, vy = 0;
let pipes = [], score = 0, alive = true;

function setup() {
  createCanvas(400, 600);
  mic = new p5.AudioIn();
  mic.start();
}

function mousePressed() { userStartAudio(); }   // allow mic

function draw() {
  background(189, 230, 255);
  let level = mic.getLevel();

  // --- physics ---
  vy += 0.4;                                    // gravity
  if (level > 0.05) vy -= level * 4;            // voice lift
  vy = constrain(vy, -7, 8);
  y += vy;

  // --- draw bird + volume bar ---
  fill(255, 183, 3);  circle(80, y, 30);
  fill(123, 63, 196); rect(10, height - 20, level * 300, 10);
```

Click first

Click the canvas once so the browser allows the microphone.

Tune

Bird too lazy? Raise 4. Too jumpy? Lower it. Too floaty? Raise gravity 0.4.

Not done

draw() continues on the next slide.

MAKE

Objective 3 · Full game · Build step 2 of 3

Pipes, crashes and score

```
  // --- pipes ---
  if (frameCount % 90 === 0) pipes.push({ x: width, gap: random(150, 450) });
  fill(42, 157, 92);
  for (let p of pipes) {
    p.x -= 3;
    rect(p.x, 0, 50, p.gap - 80);              // top pipe
    rect(p.x, p.gap + 80, 50, height);         // bottom pipe
    let inX = 80 + 15 > p.x && 80 - 15 < p.x + 50;
    let outGap = y - 15 < p.gap - 80 || y + 15 > p.gap + 80;
    if (inX && outGap) alive = false;          // crash!
    if (!p.scored && p.x + 50 < 80) { score++; p.scored = true; }
  }
  pipes = pipes.filter(p => p.x > -60);        // tidy up
  if (y > height || y < 0) alive = false;

  fill(0); textSize(28); text("Score " + score, 10, 35);
  if (!alive) {
    textSize(40); text("GAME OVER", 90, 300);
    noLoop();                                  // stop the game
  }
}
```

Gap

The gap is 160 px tall (80 above and below p.gap).

Collision

inX: bird overlaps the pipe sideways. outGap: bird is above or below the gap.

Filter

Keeps only pipes still on screen.

MAKE

Objective 3 · Build step 3 of 3

Polish and playtest

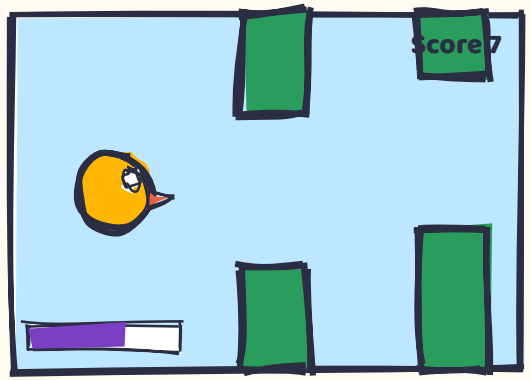
a Add a **restart**: press R to reset y, vy, pipes, score, alive, then loop().

b Add a **start screen**: 'Hum to start!' until the level first goes above 0.1.

c Make it **harder over time**: pipes move faster every 5 points.

d Draw your own bird with shapes, or load an image.

e **Playtest**: a classmate plays 3 games using only their voice. Note what confused them.



Check: A classmate can play, crash and restart without help.

Test like an engineer

Playtest notes

Test	Pass?	Fix
Bird rises when I hum		
Bird falls when quiet		
Crash on pipe = game over		
Score counts each pipe		
Classmate's best score / comment		

Debug checklist

Bird never moves up? Click the canvas first; check the browser asked for mic permission.
Always flying? Background noise: raise the 0.05 threshold.
Instant game over? The bird starts outside the canvas: y = 300 at start.

Watch players, don't explain. Every time they get confused is a clue for your next fix.

MAKE

Level up!

Game designer challenges

Choose your challenge

Pitch control: use p5.FFT to fly with HIGH notes and dive with LOW notes.

High score: save the best score with storeItem() so it survives a refresh.

Accessible mode: an easy mode with bigger gaps and slower pipes.

Hardware: use an Arduino sound sensor via Web Serial as the controller.

Think like an engineer

Why might the same settings feel easy for one player and impossible for another?

Boss level

Add a calibration screen: measure each player's quiet and loud levels, then set the threshold automatically.

Objective check

Can I really do it? Prove it.

01 What number does mic.getLevel() give for silence and for humming?

02 Point to gravity, velocity and lift in your code. What does each do?

03 Show a classmate playing. What score did they get with only their voice?

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Arcade: play 3 classmates' games. Vote for 'most fun' and 'most accessible'.

Show what you know

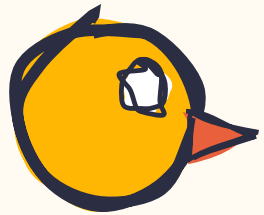
What did we learn?

- [x] Microphones turn vibrations into a level 0 to 1
- [x] Gravity changes velocity; velocity changes position
- [x] Voice above a threshold gives lift
- [x] Collisions: overlap in x AND outside the gap
- [x] Playtesting finds problems you can't see yourself

Exit ticket: what happens to vy every frame, even when you are humming?

At home

Teach someone at home to play your game using the share link.



Next class:
Python Weather API Dashboard!